PH3244 — PHYSICS LABORATORY - IV

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Experiment No. 1

Junction Field Effect Transistor

Date: 6 February, 2026

1.1 Aim

The aim of the experiment is to study the drain and transfer characterstics of an p-channel Junction Field Effect Transistor. We shall forward-bias the gate-source junction, reverse-bias the drain-source junction, and use a fixed voltage bias method to construct our setup.

1.2 Theory

A Junction Field Effect Transistor (JFET) is a voltage-controlled semiconductor device in which the drain current (I_D) is controlled by the gate-to-source voltage (V_{GS}). The gate is reverse biased, resulting in very high input impedance.

Drain Characterstics

Drain characteristics represent the variation of drain current (I_D) with drain-to- source voltage (V_{DS}) for different constant values of gate-to-source voltage (V_{GS}).

Transfer Characterstics

Transfer characteristics represent the variation of drain current (I_D) with gate-to-source voltage (V_{GS}) at a constant drain-to-source voltage (V_{DS}). These characteristics are useful for determining the transconductance of the JFET.

1.3 Experimental Observation

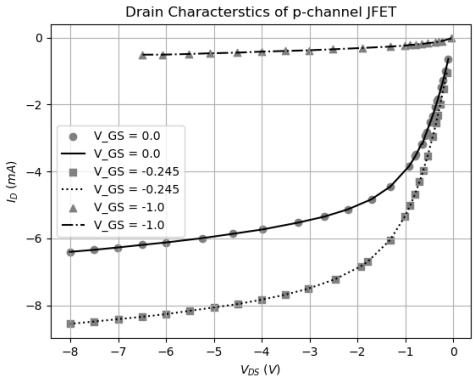
Table 1.1: Combined Drain Characteristics Observation Table

V_{DS} (V)	I_D (mA) [$V_{GS} = 0$]	V_{DS} (V)	I_D (mA) [$V_{GS} = -0.245$]	V_{DS} (V)	I_D (mA) [$V_{GS} = -1.0$]
0.10	0.64	0.13	1.05	0.03	0.02
0.16	0.99	0.20	1.53	0.23	0.10
0.21	1.29	0.27	1.99	0.30	0.12
0.24	1.47	0.32	2.32	0.37	0.14
0.32	1.85	0.36	2.56	0.54	0.17
0.33	1.90	0.42	2.95	0.66	0.19
0.37	2.07	0.53	3.53	0.78	0.21
0.43	2.34	0.63	3.98	0.90	0.22
0.47	2.52	0.70	4.31	1.01	0.24
0.56	2.83	0.80	4.69	1.32	0.27
0.58	2.92	0.90	5.02	1.89	0.31
0.64	3.18	1.01	5.35	2.51	0.35
0.66	3.18	1.31	6.04	3.02	0.38
0.78	3.50	1.78	6.69	3.51	0.40
0.79	3.55	1.93	6.83	4.00	0.42
0.92	3.84	2.46	7.21	4.51	0.45
1.31	4.45	3.03	7.50	5.08	0.47
1.71	4.83	3.51	7.68	5.52	0.49
2.20	5.13	4.01	7.83	6.07	0.51
2.70	5.35	4.50	7.96	6.49	0.51
3.25	5.53	5.00	8.06	NaN	NaN
3.98	5.73	5.50	8.16	NaN	NaN
4.60	5.86	6.00	8.26	NaN	NaN
5.24	5.99	6.50	8.34	NaN	NaN
6.00	6.12	7.00	8.41	NaN	NaN
6.49	6.19	7.50	8.48	NaN	NaN
7.01	6.27	8.00	8.55	NaN	NaN
7.51	6.34	NaN	NaN	NaN	NaN
8.01	6.40	NaN	NaN	NaN	NaN

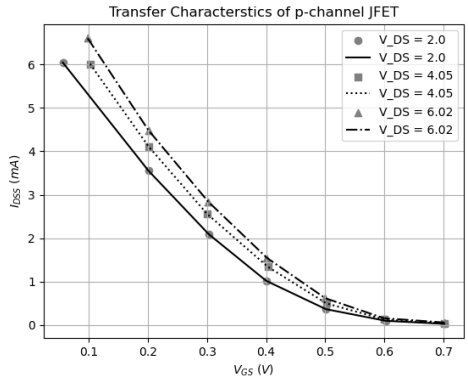
Table 1.2: Combined Transfer Characteristics Observation Table

V_{GS} (V)	I_D (mA) [$V_{DS} = 2.0$]	V_{GS} (V)	I_D (mA) [$V_{DS} = 4.05$]	V_{GS} (V)	I_D (mA) [$V_{DS} = 6.02$]
0.06	6.04	0.10	6.01	0.10	6.60
0.20	3.55	0.20	4.11	0.20	4.48
0.30	2.09	0.30	2.56	0.30	2.84
0.40	1.02	0.40	1.34	0.40	1.55
0.50	0.37	0.50	0.50	0.50	0.62
0.60	0.10	0.60	0.13	0.60	0.16
0.70	0.03	0.70	0.04	0.70	0.06

1.4 Analysis



(a) Drain Charactertics
V_{DS} vs. I_D at $V_{GS} = \{0.0, -0.245, -1.0\}$V



(b) Transfer Characterstics
V_{GS} vs. I_D at $V_{DS} = \{2.0, 4.05, 6.02\}$V

We start by plotting the characterstics curve from the observation data. We get the expected curves for both the cases as seen in Figure 1.1a and Figure 1.1b for Drain and Transfer characterstics respectively.

For the Figure 1.1a, The x-axis represents the drain-to-source voltage (V_{DS}). For a p-channel device, this value is negative. At low values of V_{DS}, the FET acts like a variable resistor. The drain current I_D rises rapidly and almost linearly as V_{DS} increases. As V_{DS} becomes more negative, the current eventually stops rising and levels off. In this region, I_D remains relatively constant regardless of changes in V_{DS}. The highest curve is when $V_{GS} = 0\,\text{V}$ (resulting in I_{DS} in the saturation region). As you make V_{GS} more positive, the entire curve shifts downward, meaning the saturation current occurs at lower levels.

For the Figure 1.1b, the control voltage (V_{GS}) is positive, and the drain current (I_D) magnitude is positive. Therefore, the curve is plotted in the first quadrant of a standard Cartesian plane. As you apply an increasingly positive voltage to the gate, the depletion region expands, narrowing the conductive channel. The specific positive voltage where the channel closes completely and I_D drops to $0\,\text{A}$ is the pinch-off voltage ($+V_P$). The curve is not a straight line; it is a parabola. The relationship is mathematically defined by Shockley's Equation.

Now, we shall move on to calculate some other parameters for the pchannel JFET like the drain resistance (r_d), transconductance (g_m) and amplification factor (μ).

First calculating the drain resistance; Drain resistance is defined as the ratio of a small change in drain-to-source voltage to the corresponding change in drain current, keeping VGS constant, when the JFET operates in the pinch-off region.

$$r_d = \left(\frac{\partial V_{DS}}{\partial I_D}\right)$$

We will find the slope for the different pinchoff regions in our drain curve.

The values of slopes and r_d from the Figure 1.2 are as follows:

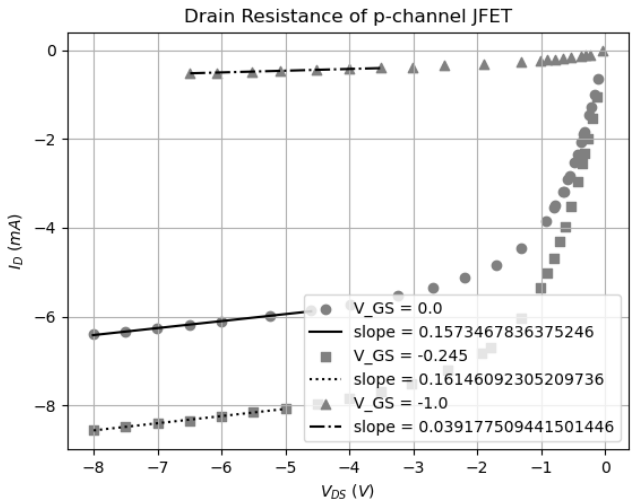


Figure 1.2: Drain Resistance

#	V_{GS}V	slope	r_dkΩ
1	0.0	$1.5735 \times 10^{-1} \pm 5.5110 \times 10^{-3}$	6.3554
2	-0.245	$1.6146 \times 10^{-1} \pm 6.1123 \times 10^{-3}$	6.1934
3	-1.0	$3.9178 \times 10^{-2} \pm 2.8449 \times 10^{-3}$	2.5525×10^{1}

We would ignore the value for r_d at $V_{GS} = -1.0$ as it doesn't seem to be in the pinchoff region properly in the observations we have taken. So, the final drain resistance we can take average for our other two readings. Thus, $r_d = 6.2744\,\mathrm{k}\Omega$.

Now, calculating the transconductance (g_m); Transconductance is defined as the ratio of a small change in drain current to the corresponding change in gate-to-source voltage, keeping VDS constant.

$$g_m = \left(\frac{\partial I_D}{\partial V_{GS}}\right)$$

I start by calculating the derivative for all the points on the transfer curve and then finding the maximum value for the transconductance for each V_{DS}.

And, the values that we have gotten are as follows:

#	V_{DS}V	g_mm℧
1	2.0	1.7172×10^{1}
2	4.05	1.9192×10^{1}
3	6.02	2.0583×10^{1}

So we can say the actual g_m will be much closer to the actual value so our estimation of g_m is 2.0583×10^{1}m℧.

Now, calculating the amplification factor; Amplification factor is defined as the ratio of a small change in drain-to-source voltage to the corresponding change in gate-to-source

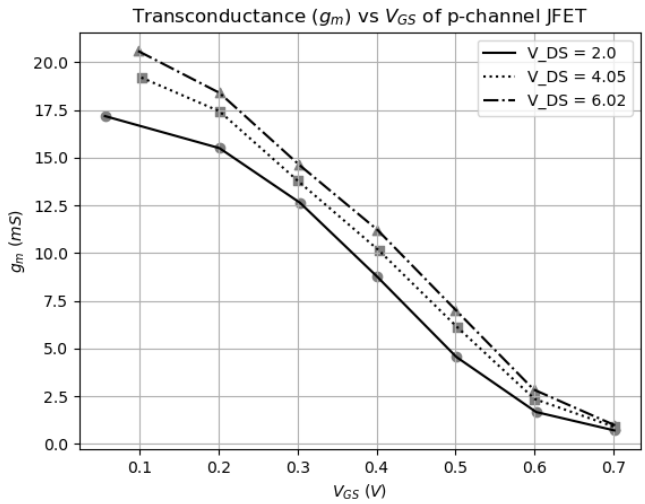


Figure 1.3: Drain Resistance

voltage, keeping ID constant.

$$\mu = \left(\frac{\partial V_{DS}}{\partial V_{GS}}\right)$$
$$\mu = r_d \times g_m$$

So the value is simply just:

$$\begin{aligned}\mu &= 6.2744 \times 2.0583 \times 10^1 \\ &= 129.1459\end{aligned}$$

1.5 Conclusion

We consider our experiment successfull as we got expected characterstics curves for the pchannel JFET and the following paramter values.

1	drain resistance (r_d)	$6.2744\,\mathrm{k}\Omega$
2	transconductance (g_m)	$2.0583 \times 10^1 \mathrm{m}\mho$
3	amplification factor (μ)	129.1459

Experiment No. 2

e/m Ratio

Date: 13 February, 2026

2.1 Aim

The aim of the experiment is to calculate the charge-by-mass (e/m) ratio of an electron, which is a fundamental constant that governs the behaviour of electrons in electromagnetic fields. Our experiment is based on Thomson's method and we use a Helmoltz coil apparatus in this setup.

2.2 Theory

This experiment works by accelerating electrons through a known potential in the tube. Some of these electrons collide with the Helium filled in the tube and radiate visible light, giving us a track of their path. The tube is placed between a pair of Helmholtz coil which provides it a uniform magnetic field. The beam is deflected in a circular path of radius r . This circular path's diameter can be measured and can be used to obtain the e/m ratio.

$$e/m = 8V/B^2d^2$$

We can simply derive this relation by the following steps:
Kinetic energy of the accelerated electron can be described as $eV = mv^2/2$. So the velocity can be written as:

$$v = \sqrt{2eV/m} \tag{2.1}$$

Passing through a uniform magnetic field B, Lorentz force acting on the electron make it go on a circular path if the field is orthogonal to the electron's trajectory. We can write this relation as the centripetal force balancing the Lorentz force, $eVB = mv^2/r$. So the velocty can be written as:

$$v = eBr/m \tag{2.2}$$

From eqns. Equation 2.1 & Equation 2.2 we obtain:

$$eBr/m = \sqrt{2eV/m} \tag{2.3}$$

$$e/m = 2V/B^2r^2 \tag{2.4}$$

For this apparatus we are controlling and fixing the magnetic field by supplying constant current to the pair of Helmholtz coils. We can write this magentic field produced by the coils as:

$$B = 2 \times \frac{\mu_0 I n}{2(5/4)^{3/2} a} = 2 \times \left(\frac{2\pi I n}{(5/4)^{3/2} a} \times 10^{-7} \right) \text{tesla}$$

where, n is the number of turns in a coil, and a is the radius of the coil.
The specifications of our specific apparatus is as follows:

Radii of Coils	14 cm
Number of turns	160 each
Accelerating Voltage	0 - 250V

Table 2.1: Specifications of the e/m Apparatus

Using all of this information, our final formulae for e/m ratio calculation is simply:

$$\frac{e}{m} = (7.576 \times 10^{-6}) \times \frac{V}{I^2 d^2} \mathrm{C\,kg^{-1}} \tag{2.5}$$

2.3 Experimental Observation

Table 2.2: First set of reading at 1 ampere

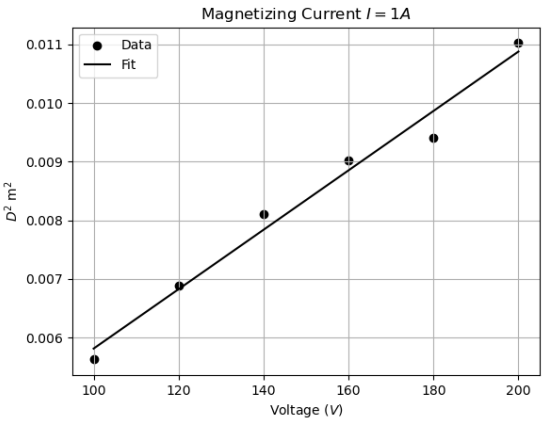
Voltage (V)	Current (A)	Left Reading (cm)	Right Reading (cm)
200	1	4.80	15.30
180	1	5.30	15.00
160	1	5.20	14.70
140	1	5.50	14.50
120	1	5.80	14.10
100	1	6.30	13.80

Table 2.3: Second set of reading at 1.5 ampere

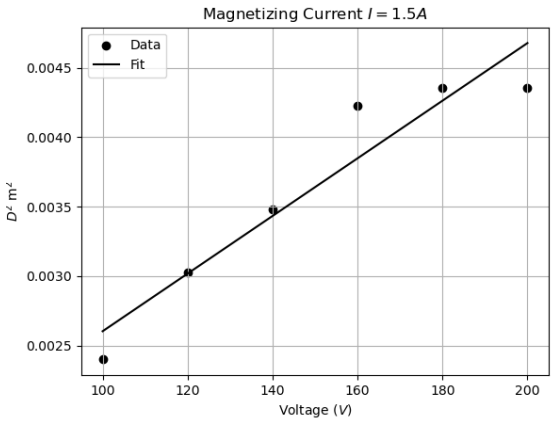
Voltage (V)	Current (A)	Left Reading (cm)	Right Reading (cm)
100	1.50	7.60	12.50
120	1.50	7.30	12.80
140	1.50	7.10	13.00
160	1.50	6.80	13.30
180	1.50	6.90	13.50
200	1.50	6.50	13.10

2.4 Analysis

Calculating from plotting



(a) D^2 vs. Voltage for $I = 1\,\mathrm{A}$



(b) D^2 vs. Voltage for $I = 1.5\,\mathrm{A}$

We have simply linearly fitted the data to find the following plots; Figure 2.1a & Figure 2.1b. For the plot Figure 2.1a the slope of the linear fit is $5.0693 \times 10^{-5} \pm 3.6199 \times 10^{-6}$ and for the plot Figure 2.1b the slope of the linear fit is $2.0731 \times 10^{-5} \pm 3.2602 \times 10^{-6}$. From the slope we can easily calculate the e/m ratio using the relation as given in Equation 2.5.
For Figure 2.1a:

$$\begin{aligned}\frac{e}{m} &= (7.576 \times 10^{-6}) \times \frac{1}{5.0693 \times 10^{-5} \times (1)^2}\mathrm{C\,kg^{-1}} \\ &= 1.4945 \times 10^{11} \pm 1.0672 \times 10^{10}\end{aligned}$$

For Figure 2.1b:

$$\begin{aligned}\frac{e}{m} &= (7.576 \times 10^{-6}) \times \frac{1}{2.0731 \times 10^{-5} \times (1.5)^2}\mathrm{C\,kg^{-1}} \\ &= 1.6242 \times 10^{11} \pm 2.5542 \times 10^{10}\end{aligned}$$

Directly Calculating from the data by taking a mean

Table 2.4: Directly calculated e/m ratio

#	Voltage (V)	Current (A)	Diameter (m2)	V/I2d2	e/m
1	200	1.00	0.10	18140.59	137 433 106 575.96
2	180	1.00	0.10	19130.62	144 933 574 237.43
3	160	1.00	0.10	17728.53	134 311 357 340.72
4	140	1.00	0.09	17283.95	130 943 209 876.54
5	120	1.00	0.08	17419.07	131 966 903 759.62
6	100	1.00	0.08	17777.78	134 684 444 444.44
7	100	1.50	0.05	18510.81	140 237 863 853.02
8	120	1.50	0.06	17630.85	133 571 349 862.26
9	140	1.50	0.06	17874.81	135 419 579 303.52
10	160	1.50	0.07	16831.03	127 511 900 065.75
11	180	1.50	0.07	18365.47	139 136 822 773.19
12	200	1.50	0.07	20406.08	154 596 469 747.98

Thus, the mean value for e/m obtained is $1.3706 \times 10^{11}$and the standard deviation as 6.8718×10^{9}.

Error Analysis

We ae getting lower than theoritcal value consistently for all our observations. This is predominantly due to the fact that it is impossible for us to accurately measure the diameter of the path of the beam. We simply took the outermost ring of the thick beam that we observed, to be consistent among our readings, but this simply lead to a higher estimation of r the radius of trajectory. And finally lead to us calculating a lower e/m ratio.

The error in our final result compared to the theoritcal value is:

$$\begin{aligned}\text{Error} &= \frac{1.7588 \times 10^{11} - 1.6242 \times 10^{11}}{1.7588 \times 10^{11}} \times 100\% \\ &= 7.65\%\end{aligned}$$

2.5 Conclusion

The theoritical value for e/m ratio is $1.76\,\mathrm{C\,kg^{-1}}$. Our best result was during the second set of calculations albeit with a higher standard deviation which was equal to $1.6242 \times 10^{11} \pm 2.5542 \times 10^{10}\mathrm{C\,kg^{-1}}$. This gave us an error percentage equal to 7.65%.

Our values were consistently below the theoritical value which has been explained above sufficiently. Overall we achieved a reasonably close value for e/m ratio and can consider the experiment as successful.